A small SF basis for SF-FSA-VI in deterministic and stochastic domains

October 19, 2026

1 Setting

Finite S, A, transitions $P(s' \mid s, a)$, $\gamma \in (0, 1)$. There are d *exits*; entering exit i emits the feature $\mathbf{e}_i$ and ends the episode, and no other feature is emitted. For $\mathbf{w} \in \mathbb{R}^d_{\geq 0}$ the reward is $r_{\mathbf{w}} = \boldsymbol{\phi} \cdot \mathbf{w}$, and a policy's action SFs $\boldsymbol{\psi}^\pi(s, a) \in [0, 1]^d$ (with $\sum_i \psi^\pi_i \leq 1$) give $Q^\pi_{\mathbf{w}} = \boldsymbol{\psi}^\pi \cdot \mathbf{w}$. SF-FSA-VI [1] plans an acyclic automaton in reverse topological order: state u gets $w_i(u) = 1$ if exit i's symbol ends the task, $w_i(u) = V_{\mathbf{w}(v)}(i)$ (the value from exit i in the next state v) if it leads to v, and 0 otherwise; the agent executes a policy for $\mathbf{w}(u)$ in u.

A *base* is a set $L = \{\pi_1, \ldots, \pi_k\}$ with known SFs; $B(s) = \{\pi_j(s)\}_j$ are its actions at s. At a weight $\mathbf{w}$:

- GPI: $\pi^G(s) = \arg\max_a \max_j Q^{\pi_j}_{\mathbf{w}}(s, a)$; Kuric et al. pass the estimate $\max_{j,a} Q^{\pi_j}_{\mathbf{w}}(i, a)$ upstream.
- metapolicy π_{meta}: the optimal policy of the MDP restricted to the actions $B(s)$ [2]; $Q_{\text{meta}}(s, a) = r_{\mathbf{w}}(s, a) + \gamma\, \mathbb{E}_{s'} V_{\text{meta}}(s')$ for every $a \in A$.
- hybrid policy $\pi^{\text{hyb}}(s) = \arg\max_{a \in A} Q_{\text{meta}}(s, a)$ [2]: one improvement step, over all actions, on top of the metapolicy.

2 Deterministic domains: the d extreme policies suffice

Theorem 1. *Let P be deterministic and π_i an optimal policy for $\mathbf{e}_i$, $i = 1, \ldots, d$. For every $\mathbf{w} \geq 0$ and every (s, a),*

$$\max_i \ \boldsymbol{\psi}^{\pi_i}(s, a) \cdot \mathbf{w} \ = \ Q^*_{\mathbf{w}}(s, a).$$

Hence GPI over $\{\pi_1, \ldots, \pi_d\}$ is optimal for every weight, its estimate is exact, the metapolicy and π^{hyb} are optimal too, and zero-shot SF-FSA-VI over the d extremes is optimal for every FSA.

Proof. Let $D_i(s, a) = Q^*_{\mathbf{e}_i}(s, a) = \psi^{\pi_i}_i(s, a)$. From (s, a) any policy generates one trajectory, which enters at most one exit, say i after t further steps, so $Q^\pi_{\mathbf{w}}(s, a) = w_i \gamma^t \leq w_i D_i(s, a)$. Hence $Q^*_{\mathbf{w}} \leq \max_i w_i D_i$. Conversely $\boldsymbol{\psi}^{\pi_i} \cdot \mathbf{w} \geq w_i \psi^{\pi_i}_i = w_i D_i$ (nonnegative features and weights), while $\boldsymbol{\psi}^{\pi_i} \cdot \mathbf{w} = Q^{\pi_i}_{\mathbf{w}} \leq Q^*_{\mathbf{w}}$. So all three are equal, GPI is greedy for $Q^*_{\mathbf{w}}$, and the exit values passed upstream are $V^*_{\mathbf{w}}$; induction over the automaton gives the FSA statement. By Proposition 1(i), $\max_j Q^{\pi_j}_{\mathbf{w}} \leq Q_{\text{meta}} \leq Q^*_{\mathbf{w}}$, so $Q_{\text{meta}} = Q^*_{\mathbf{w}}$ and π^{hyb} is optimal; and $B(s)$ contains an optimal action, $\pi_{i^*}(s)$ with $i^* = \arg\max_i w_i D_i(s, \pi_i(s))$, so the metapolicy is optimal too. □

The only step that fails with stochastic P is $Q^\pi_{\mathbf{w}} \leq \max_i w_i D_i$: a policy can split its probability between exits (keep both open), and then $Q^*_{\mathbf{w}} > \max_i w_i D_i$. Numerically, the certificate of [3] confirms the theorem on deterministic Office: every weight of the 6-exit simplex, $\Gamma = 5.7 \cdot 10^{-8}$, with the 6 extremes.

3 Stochastic domains: SF-FSA-VI over a small base

Proposition 1. *For any base and $\mathbf{w}$: (i) $\max_j Q^{\pi_j}_{\mathbf{w}} \leq Q_{\text{meta}} \leq Q^{\pi^{\text{hyb}}}_{\mathbf{w}}$ and $V^{\pi^{\text{hyb}}}_{\mathbf{w}} \geq V_{\text{meta}} \geq \max_j V^{\pi_j}_{\mathbf{w}}$: the hybrid's guaranteed value dominates GPI's. (ii) Let $\Delta(s)$ be the gap between $V^*_{\mathbf{w}}(s)$ and the best strictly suboptimal action value at s. If $\gamma \|V^*_{\mathbf{w}} - V_{\text{meta}}\|_\infty < \min_s \Delta(s)$, then π^{hyb} is optimal; the same argument for GPI needs $\gamma \min_j \|V^*_{\mathbf{w}} - V^{\pi_j}_{\mathbf{w}}\|_\infty < \min_s \Delta(s)$, which is stronger by (i). (iii) If some optimal policy for $\mathbf{w}$ takes only actions in $B(s)$, then π_{meta} and π^{hyb} are optimal.*

Proof. (i) Each π_j is feasible in the restricted MDP, so $V_{\text{meta}} \geq V^{\pi_j}$ and $Q_{\text{meta}} = r + \gamma P V_{\text{meta}} \geq r + \gamma P V^{\pi_j} = Q^{\pi_j}$. π^{hyb} is greedy for Q_{meta}, so $T^{\pi^{\text{hyb}}} V_{\text{meta}} \geq V_{\text{meta}}$ and, by monotonicity, $V^{\pi^{\text{hyb}}} \geq V_{\text{meta}}$.

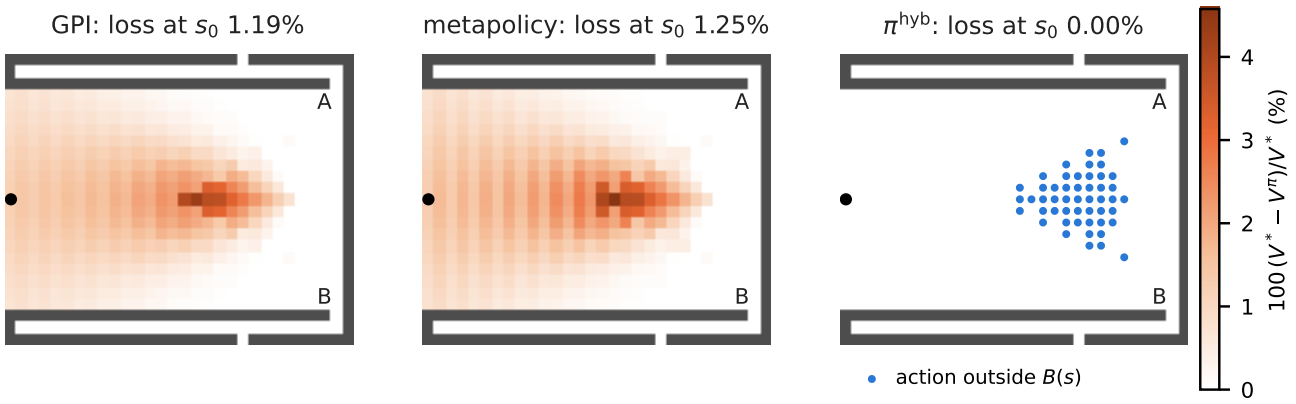


Figure 1: Why the metapolicy: automaton state u_0 of the windy task "(A or B), then a corridor exit", where both exits are worth the same, $\mathbf{w}(u_0) = (0.63, 0.63, 0, 0, 0, 0)$. Colour: loss $V^* - V^\pi$ of each policy built from the 6 extremes, relative to V^*, at every state of the windy zone (start •, exits A, B on the right). π^{hyb} takes the action RIGHT, which no extreme takes, at the blue states.

(ii) $0 \le Q^* - Q_{\text{meta}} = \gamma P(V^* - V_{\text{meta}}) \le \gamma \|V^* - V_{\text{meta}}\|_\infty$, so an optimal action scores above every action that is suboptimal by Δ; for GPI, $Q^*(s,a) - \max_j Q^{\pi_j}(s,a) \le \gamma \min_j \|V^* - V^{\pi_j}\|_\infty$. (iii) The restricted MDP then contains an optimal policy, so $V_{\text{meta}} = V^*_{\mathbf{w}}$. □

(i) bounds what π^{hyb} guarantees; GPI's executed value can occasionally be higher [2]. Neither (ii) nor (iii) gives a number of base policies in terms of d: if the optimal behaviour needs two or more consecutive actions that no base policy takes, one improvement step is not enough.

Why the metapolicy: hedging. In the windy two-stage domain [3] the wind decides whether the agent reaches exit A or B, and from each of them corridors lead to further exits. In a task such as "(A or B), then a corridor exit" both exits are worth the same in u_0 (Figure 1). The optimal policy hedges: it moves forward without pushing up or down, and commits only when the wind has decided. Each extreme pushes towards its own exit from the first step. GPI evaluates every action followed by a *single* extreme, so it never values waiting, and loses 1.19% at the start. The metapolicy switches between the extremes at every state ("up if the wind pushed us up"), but each of them still pushes, so it loses 1.25%. π^{hyb} evaluates each action followed by the *metapolicy*: the one action outside $B(s)$ that matters, RIGHT (advance without choosing a side), is now valued with a continuation that commits later, and π^{hyb} takes it at 50 central states. Its loss is 0 at the start and at most 0.04% anywhere. The metapolicy is therefore not useful as a policy but as the *continuation* that makes one non-base action enough, which a single base policy is not.

Certificate. A policy is optimal for $\mathbf{w}$ iff its Bellman gap $\max_a [r_{\mathbf{w}} + \gamma P V^\pi](s,a) - V^\pi(s)$ is zero at every state. Checking it costs one evaluation of π and one lookahead (prediction, no control), and policy iteration detects it when it stops. Algorithm 1 tries the cheap candidates first and only then improves. By induction over the automaton, the executed policy is optimal and the planned values are exact. Evaluations are iterative and warm-started at a sub-solution of the evaluated policy (the GPI estimate, $\max_j V_j$, V_{meta}), so no linear system is solved. *Without the model*, rungs 2–3 become one Q-learner at $\mathbf{w}(u)$, initialised at the GPI estimate, with acting and bootstrapping restricted to $B(s)$ (the metapolicy) and every entry keeping the visit count of the base entry it was copied from (*inherited counts*); otherwise the first noisy updates overwrite an initialisation that is already near-optimal. The base itself is built once with Certify-or-Grow [2], or is just the d extremes.

Algorithm 1 SF-FSA-VI over a base L in a stochastic domain

for u in reverse topological order **do**
  $\mathbf{w}(u) \leftarrow$ exit weights: 1, $V_v(i)$ (value of the policy executed in v) or 0
  $\pi \leftarrow$ GPI over L at $\mathbf{w}(u)$; $V \leftarrow$ evaluate π ▷ rung 1: zero-shot
  if Bellman gap of (π, V) is nonzero **then**
    $\pi \leftarrow \pi^{\text{hyb}}$ at $\mathbf{w}(u)$; $V \leftarrow$ evaluate π ▷ rung 2: metapolicy, then π^{hyb}
    while Bellman gap of (π, V) is nonzero **do** ▷ rung 3: improve
      $\pi \leftarrow$ greedy for $r_{\mathbf{w}} + \gamma PV$; $V \leftarrow$ evaluate π
    end while
  end if
  execute π in u; $V_u \leftarrow V$
end for

Table 1: Algorithm 1 with the model, per task type (number of tasks in brackets), base = the 6 extremes. Max regret over the tasks, relative to the optimal value, of zero-shot GPI [1], the metapolicy and π^{hyb} executed without certification; share of automaton states where Algorithm 1 stops at rung 1/2/3; average cost per task of the certified optimum in full sweeps ($|S||A|$ one-step backups; an evaluation sweep counts $1/|A|$): value iteration from scratch (best tolerance, then policy iteration to zero gap), GPI followed directly by policy iteration (rung 2 skipped), and Algorithm 1. $|U|$: automaton states per task. Every certified plan has regret 0.

			max relative regret, uncertified			rungs (%)	sweeps per task, certified		
domain	task type	$\lvert U\rvert$	GPI	metapolicy	π^{hyb}	1/2/3	VI	GPI + PI	Alg. 1
Office	sequential (1)	3.0	$6.4 \cdot 10^{-9}$	0	0	33/67/0	109	62	84
$p = 0.1$	disjunction (1)	2.0	$9.3 \cdot 10^{-6}$	$1.2 \cdot 10^{-7}$	0	0/50/50	66	50	77
	composite (1)	4.0	$1.8 \cdot 10^{-5}$	$9.2 \cdot 10^{-6}$	$2.5 \cdot 10^{-6}$	25/50/25	147	94	154
	random (100)	1.7	$1.8 \cdot 10^{-5}$	$2.1 \cdot 10^{-5}$	$1.7 \cdot 10^{-5}$	7/85/8	58	32	52
Office	sequential (1)	3.0	$4.9 \cdot 10^{-6}$	0	0	0/100/0	234	140	230
$p = 0.3$	disjunction (1)	2.0	$9.9 \cdot 10^{-5}$	$5.3 \cdot 10^{-5}$	$3.5 \cdot 10^{-8}$	0/50/50	144	95	175
	composite (1)	4.0	$3.3 \cdot 10^{-4}$	$8.4 \cdot 10^{-4}$	0	0/100/0	282	190	348
	random (100)	1.7	$1.3 \cdot 10^{-3}$	$1.6 \cdot 10^{-4}$	$1.2 \cdot 10^{-6}$	19/76/5	117	66	111
Windy	A or B, then corr. (52)	3.0	$1.2 \cdot 10^{-2}$	$1.2 \cdot 10^{-2}$	0	90/2/8	102	12	19
	A only, then corr. (8)	2.0	0	0	0	100/0/0	42	2	2

4 Results

Domains: double slit (wind ± 3, $\gamma = 0.9$), Office with slip p (the chosen action is replaced by one of the other three w.p. p, $\gamma = 0.99$) and windy two-stage ($\gamma = 0.95$) [3]. Office tasks: the 3 of [1], *sequential* (C, then M, then O), *disjunction* (C or M, then O) and *composite* (C then M, or M then C, then O), plus 100 random acyclic automata (up to 4 states, 1–2 symbols per state). Windy: 60 tasks "(A or B), then 1–2 corridor exits" (52; u_0 hedges) or "A (or B), then corridor exits" (8). The base is always the $d = 6$ extremes.

- **Where the metapolicy matters (Table 1, Figures 1 and 2).** Where GPI is already near-optimal there is nothing to fix: in Office with $p = 0.1$ every policy is within $2 \cdot 10^{-5}$. Where it is not, π^{hyb} fixes it: in Office with $p = 0.3$, 9 of the 103 tasks lose more than 10^{-4} with GPI (up to 0.13%); the metapolicy brings 8 of them below 10^{-4} and π^{hyb} makes all 9 exact. In windy, 12 of the 52 hedging tasks lose more than 10^{-4} (up to 1.2%); the metapolicy fixes none of them (it cannot hedge, Figure 1) and π^{hyb} is exact on all 60. Over held-out weights (Figure 2), GPI over the 6 extremes is optimal at 86%, 69% and 93% of them and π^{hyb} at 97%, 93% and 96%; in double slit 9 policies make π^{hyb} optimal everywhere, where the CCS has 181.

Table 2: Without the model: Office with slip 0.05, base = the 6 extremes learned by SF Q-learning (2.5M steps each, visit counts kept). Millions of environment steps per task until the greedy policy of every automaton state is within 10^{-3} of the optimum at its entry states (checked by an oracle; budget 2M per automaton state, a state that does not get there counts 2M). All learners are scalar Q-learning at $\mathbf{w}(u)$ with step $1/(n+1)^{0.7}$; 20 tasks (24 distinct automaton states), 2 seeds; $|U|$: automaton states per task.

			GPI start, all actions		GPI start, metapolicy ($B(s)$)	
task type	$\|U\|$	from scratch	fresh counts	inherited counts	fresh counts	inherited counts
sequential	3.0	2.47	2.42	1.63	2.41	**0.59**
disjunction	2.0	2.31	2.10	1.65	1.93	**1.40**
composite	4.0	4.10	3.43	1.89	3.64	**1.56**
random (17)	1.5	0.75	0.58	**0.38**	0.69	0.39
all 20 tasks		1.08	0.89	0.58	0.99	**0.51**

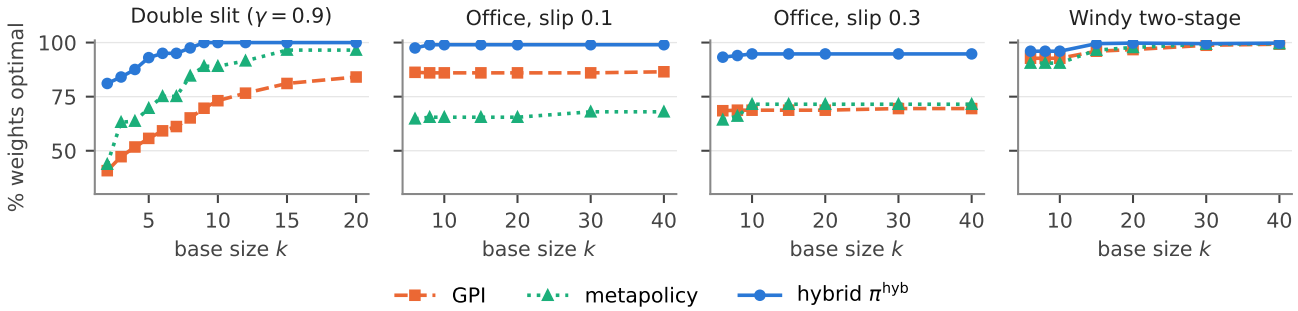


Figure 2: Share of held-out weights where each policy is optimal at the start and every exit, with the first k policies found by OLS as base (the first d are the extremes). Policy iteration from π^{hyb} reaches the optimum in at most 3 steps (mean ≤ 0.3) in every case.

Table 3: Building the ϵ-CCS from the base (the d extremes), with the model: average cost per CCS policy beyond the base in SF sweeps ($|S||A|$ backups of d-dimensional SFs) and, in brackets, the policies trained beyond the base.

	Double slit, exact	Windy, 10^{-5}	Office 0.1, $2 \cdot 10^{-3}$	Office 0.3, $5 \cdot 10^{-3}$
CCS policies beyond the base	308	111	256	232
SFOLS, learned from scratch	889 (308)	471 (111)	–	–
SFOLS, GPI warm start	20.7 (307)	47.4 (111)	57.2 (256)	143.6 (232)
Certify-or-Grow	21.3 (11)	51.0 (15)	60.0 (18)	150.6 (16)

- **Cost of the certified optimum, with the model (Table 1).** Every task is solved exactly. The cheapest route is GPI followed directly by policy iteration: 1.8× cheaper than value iteration from scratch in Office and 8.6× in windy (one evaluation certifies GPI in 90% of the windy automaton states). With rung 2, π^{hyb} is certified at 76–85% of the Office automaton states, but its restricted policy iteration costs more than the 2 improvement steps it saves, so Algorithm 1 is 1.5–1.7× more expensive than GPI + PI (about the cost of value iteration in Office). With the model, rung 2 buys a near-optimal policy before any improvement step over all actions, not speed. GPI rarely passes the certificate in Office (8–18%) although its regret is tiny: the gap must vanish at *every* state, also far from the task's entry states.
- **Without the model (Table 2).** Zero-shot GPI over the learned extremes is already within 10^{-3} at 22 of the 24 automaton states, so the learner's job is mainly not to spoil it. With fresh counts it does: two thirds of the runs that start optimal drift away, which costs 0.44M steps per automaton

state on average. Inherited counts cut the cost per task 1.5–1.9×, and the metapolicy (restricting the learner to $B(s)$) helps most on the longer tasks: 0.59M instead of 1.63M on the sequential task and 1.56M instead of 1.89M on the composite one, 0.51M instead of 0.58M over all 20 tasks, 2.1× fewer than learning from scratch (one learned base, 2 seeds: indicative).

- **The CCS from the base (Table 3).** With the model, Certify-or-Grow trains 11–18 policies instead of 111–308, at about the cost of warm-started SFOLS (+3–8% sweeps) and 9–42× less than SFOLS from scratch. Without the model (Figure 1 of [2] with SF Q-learning, Certify-or-Grow warm-starting its base like SFOLS, 3 seeds) there is no reliable gain: in double slit (wind ±1, $\epsilon = 10^{-3}$) the metapolicies converge in about 0 steps after the base, but the base is 80–100% of the cost (1.14× fewer steps to converge than warm SFOLS, 1.05× model-free); in Office with slip 0.05 ($\epsilon = 0.05$), each metapolicy still needs about 1M steps, Certify-or-Grow with the exact certificate falls back 14–23 times and needs 150M steps against 129M, and the model-free certificate needs 112M but accepts 14–18 metapolicies whose true gap exceeds the tolerance.
- **Limits.** Rung 3 needs the model or a learner; certifying without the model is the open part (sampled completion with 100 samples per pair wrongly certified 15–22% of the metapolicies). The zero-gap certificate is stricter than SF-FSA-VI needs; a certificate weighted by reachability from the entry states would let GPI stop at rung 1 far more often.

References

[1] D. Kuric, G. Infante, V. Gómez, A. Jonsson, H. van Hoof. Planning with a learned policy basis to optimally solve complex tasks. ICAPS 2024.

[2] Certify-or-Grow, `paper/article.tex`.

[3] Certified SF libraries, `notes/certified_library.tex`.